

Midterm 1 Part a

19 November 2024

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I. QUESTIONS (39 POINTS)

1. No ... theorem that says there is no such thing as the best learning algorithm?
 - a) Late Breakfast
 - b) Free Lunch
 - c) Spoon
 - d) Last Supper
2. If the classifier is trained more than it was needed and it memorizes its decisions, then the classifier is ...?
 - a) Overfitting
 - b) Underfitting
 - c) Fitting
3. If the classifier has large test error but less train error, then the classifier is ...?
 - a) Overfitting
 - b) Underfitting
 - c) Fitting
4. Which of the following experiment design techniques has the highest space complexity?
 - a) Best guess
 - b) One factor at a time
 - c) Factorial design
5. Which of the following algorithms has the highest space complexity?
 - a) K nearest neighbor
 - b) Qda
 - c) Lda
 - d) Nearest Mean
 - e) Naive Bayes
6. Which of the following algorithms has no training complexity?
 - a) K nearest neighbor
 - b) Qda
 - c) Lda
 - d) Nearest Mean
 - e) Naive Bayes
7. Which of the following classifiers is not linear?
 - a) Qda
 - b) Lda
 - c) Nearest Mean
 - d) Naive Bayes
8. What will be the training test percentage if we do 10-fold cross-validation?
 - a) 10 percent
 - b) 20 percent
 - c) 50 percent
 - d) 80 percent
 - e) 90 percent
9. Which of the following datasets is used to get the performance of the classifier in real life?
 - a) Train set
 - b) Test set
 - c) Validation Set
10. What will be the testing test percentage if we do 10-fold cross-validation?
 - a) 10 percent
 - b) 20 percent
 - c) 50 percent
 - d) 80 percent
 - e) 90 percent
11. Which of the following datasets is used to learn the classifier?
 - a) Train set
 - b) Test set
 - c) Validation Set
12. Which of the following classifier finds the mean of each class and assigns the instance to the nearest of them?
 - a) Qda
 - b) Lda
 - c) Nearest Mean
 - d) Naive Bayes
13. Which of the following classifiers does not make any dependency assumption between features?
 - a) Qda
 - b) Lda
 - c) Nearest Mean
 - d) Naive Bayes

14. Which of the following classifiers is the most simple one?

- a) Qda
- b) Lda
- c) Nearest Mean
- d) Naive Bayes

15. What is the square of standard deviation?

- a) mean
- b) variance
- c) gauss
- d) bias

16. At which points does the normal distribution take its peak value?

- a) mean
- b) variance
- c) gauss
- d) bias

17. When the covariance of two variables are zero and their variances are equal, what is the shape of the isoprobability contour plot?

- a) square
- b) ellipse
- c) circle
- d) ellipse with 45 degree

18. When the covariance of two variables are zero and their variances are different, what is the shape of the isoprobability contour plot?

- a) square
- b) ellipse
- c) circle
- d) ellipse with 45 degree

19. If we have a $K = 4$ class classification dataset, how many one-vs-one 2 class problems can be defined for this dataset?

- a) 4
- b) 5
- c) 6
- d) 8
- e) 16

20. If we have a $K = 4$ class classification dataset, how many one-vs-rest 2 class problems can be defined for this dataset?

- a) 4
- b) 5
- c) 6
- d) 8
- e) 16

21. In which of the following machine learning applications there is a supervisor and it tries to estimate a continuous entity from a set of possibilities?

- a) Association
- b) Classification
- c) Regression
- d) Unsupervised Learning
- e) Reinforcement Learning

22. In which of the following machine learning applications there is no supervisor?

- a) Association
- b) Classification
- c) Regression
- d) Unsupervised Learning
- e) Reinforcement Learning

23. Which of the following is not a dimension of a machine learning algorithm?

- a) Model
- b) Loss function
- c) Optimization procedure
- d) Variance

Assume you wish to use the K-nearest neighbors algorithm on this dataset.

A	B	C	D	Output
T	T	T	T	F
F	F	T	F	T
F	T	F	T	F
T	T	F	F	F
T	F	T	T	F
F	F	F	F	?

24. Which class will be assigned to the test instance (last instance) if $k = 1$?

- a) T
- b) F

25. Which class will be assigned to the test instance (last instance) if $k = 3$?

- a) T
- b) F

$$\mathcal{X}_1 = \begin{bmatrix} x_1 & x_2 & x_3 & C \\ x^{(1)} & 4 & 2 & green & yes \\ x^{(2)} & 6 & ? & yellow & no \\ x^{(3)} & 7 & 2 & blue & yes \\ x^{(4)} & 3 & 1 & green & no \\ x^{(5)} & 3 & 1 & blue & no \end{bmatrix}$$

26. If feature 2 of $x^{(2)}$ is missing, what will be the imputation value for it?

- a) 1
- b) 1.2
- c) 1.3
- d) 1.5
- e) 2